Topic -4:-

Attributes & Methods in python

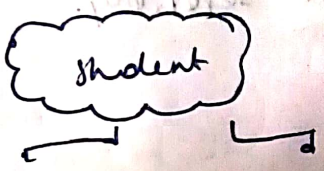
* Attributes are variables that store data inside a class
* Methods are function defined inside a class that describes the behavior of object

Attributes :- "what the object has"

Methods :- "what the object does"

So every object contains:-

* Data (attributes)
* Action (methods)



Attributes (data)	Methods (behavior)
* name	* study()
* age	* write_exam()
* Marks	* display_info(),

Types of Attribute

* class Attributes
* instance attributes.

① class Attributes

These are share by all objects

All object share the same value

because it belongs to the class.

not to individual objects.

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Ex

```
class student:
    school = "SVCE"

s1 = student()
s2 = student()

Print(s1.school)
Print(s2.school)
```

② Instance Attributes

These belong to each object seperately.

```
class student:
    def __init__(self, name, age):
        self.name = name # instance attribute
        self.age = age # instance attribute

s1 = student("Nithesh", 20)
s2 = student("gagan", 26)

Print(s1.name, s1.age)
Print(s2.name, s2.age)
```

Here

s1 and s2 have different data

Each object stores its own value.

Methods in python:

method are function defined inside a class

* Every method must have self as the first parameter..

example

```
class Student
    def __init__(self, name):
        self.name = name

    def greet(self):
        print("Hello my name is", self.name)

s1 = Student("Nitheesh")
s1.greet()
```

Hello my name is Nitheesh

** self

self represent current object

self → s1 //